EXCELLENCE

CLASS 11 - 2026-27

Physics | Units 03

Units and Measurement

Dimensional Applications and Significant Figures

By - Excellence Programme

8 October 2026

Topics to be covered

01 From Units 02: derive a formula (pendulum)

02 Uses of dimensional analysis

03 sin / log / exp — find unknown constants

04 New fundamental quantities (F, a, t …)

05 Deducing relations by matching powers

06 Dimensional vs dimensionless constants

07 Limitations + significant figures

Where we left Units 02

Last class we locked homogeneity: things that are added must match in dimensions. We also treated sin θ, e^x and log x as the number 1 for dimensions.

Units 02 ended by opening the next idea — slide 51: use dimensions to find the powers in a formula. We finish that idea first, then go into exam applications and significant figures.

Start today from Units 02 slide 51: derive the pendulum formula

Class notes will be saved after teaching. This deck is for the board.

Using dimensions to derive a formula

Sometimes we know which quantities a formula depends on, but we do not know the powers. Dimensional analysis can find those powers. We assume a product form, write dimensions on both sides, and match the powers of M, L and T one by one.

Example idea: time period T of a simple pendulum depends on length l and acceleration due to gravity g. We write $T = k\, l^a g^b$, where k is a pure number. Then we find a and b. The method cannot give the value of k — experiment or deeper theory gives $k = 2\pi$.

$T \propto l^a g^b$ → find a and b by comparing powers

What we assume when we derive

We assume the unknown relation is a product of powers: quantity = constant × (first quantity)^a × (second)^b × …. This works when each variable appears once as a power. It fails if the true law has a sum of two different terms, or a trigonometric factor that we forgot to treat as dimensionless.

Also, if a quantity depends on more variables than we can balance with M, L, T alone, dimensions may not give a unique answer. Then we need more physics. Keep that honesty for board answers under limitations.

k is a pure number here. Dimensions ignore it.

Rough work / extra work

Pendulum step by step

Write $T = k\, l^a\, g^b$. Now match dimensions of both sides.

Left side $[T] = T^1$. Right side $= L^a \times (L\,T^{-2})^b$

Right side becomes $L^{a+b}\,T^{-2b}$

Match time power: $1 = -2b \Rightarrow b = -1/2$

Match length power: $0 = a + b \Rightarrow a = +1/2$

So $T = k\,\sqrt{(l/g)}$. Real formula uses $k = 2\pi$.

Uses of dimensional analysis

Dimensional analysis has three standard school uses. First, check whether an equation is dimensionally correct. Second, convert a numerical value from one system of units to another using n1 u1 = n2 u2. Third, find how physical quantities are related — the powers in a formula — when we know which quantities matter.

For a 3-mark CBSE answer, write the definition of homogeneity briefly, list these three uses, and add one limitation. That structure scores cleanly without long English.

Check · Convert · Derive

Uses of dimensional analysis (3 marks)

1. To check whether an equation is dimensionally correct.

2. To convert a value from one system of units to another.

3. To find the relation between physical quantities (derive formula).

Check · Convert · Derive

Limitation: cannot find dimensionless constants like 2π or 1/2.

NEET: time period from α and β

Max acceleration of oscillating particle is α, max velocity is β. Time period is

1. $2\pi\ \alpha/\beta$
2. $2\pi\ \alpha\ \beta$
3. $2\pi\ \beta/\alpha$
4. $2\pi\ \alpha^2\ /\ \beta^2$

Solution: NEET: time period from α and β

Easy way: check which option has dimension of time.

Step 1: In SHM, max speed $\beta = A\omega$, max acceleration $\alpha = A\omega^2$

Step 2: Divide: $\alpha/\beta = \omega$. Time period is $T = 2\pi / \omega$

Step 3: So $T = 2\pi \times (\beta/\alpha)$

Final answer: $2\pi\ \beta/\alpha$ (option 3)

Rough work / extra work

Which does NOT have dimension of time?

L, C, R are inductance, capacitance, resistance. Which has no dimension of time?

1. RC
2. L/R
3. √(LC)
4. L/C

Solution: Which does NOT have dimension of time?

Remember three time formulas from circuits.

Step 1: RC has dimension of time. L/R has dimension of time.

Step 2: √(LC) also has dimension of time.

Step 3: L/C does not become time. That is the odd one out.

Final answer: L/C (option 4)

Inside sin, log and e^x — the hard rule

Any argument sitting inside sin, cos, tan, log or e must be dimensionless. That means its dimensions are $M^0\ L^0\ T^0$. If two pieces are added inside that argument, each piece must also be dimensionless — and they must match each other.

Outside, the whole sin or log is just a pure number for dimensions. So if Force = A cos(Bx), then [A] = [Force], and [Bx] = 1.

Inside = dimensionless · Outside multiplier matches the left side

How to attack such an equation

Step 1: Mark every + or − inside the sin / log / exp. Those pieces match.

Step 2: Each piece alone must be dimensionless (equal to 1 for dimensions).

Step 3: The factor in front of sin / log / exp matches the left-hand quantity.

Step 4: Multiply or divide to get the combination the question asks for.

Do not invent physics. Read only what the equation allows.

If λt sits inside sin, and λ is length, that equation is dimensionally wrong.

Force with cos and sin

F = A cos(Bx) + C sin(Dt). Find dimensions of (A D)/B.

Solution: Force with cos and sin

Idea: Bx and Dt are dimensionless. A and C carry the force.

Step 1: $[Bx] = 1 \Rightarrow [B] = 1/[x] = L^{-1}$. $[Dt] = 1 \Rightarrow [D] = T^{-1}$

Step 2: $[A] = [C] = [F] = M\,L\,T^{-2}$

Step 3: $[A\,D\,/\,B] = [F] \times T^{-1} / L^{-1} = [F] \times (L\,/\,T)$

Final answer: same as Force × length / time

Rough work / extra work

Velocity with square-root density

V = α / (β + √density). Find [α] and [β].

Solution: Velocity with square-root density

Idea: β is added to √density, so they match. Then α / that piece = velocity.

Step 1: $[\beta] = [\sqrt{\text{density}}] = (M\,L^{-3})^{1/2} = M^{1/2}\,L^{-3/2}$

Step 2: $[\alpha] = [V] \times [\beta] = L\,T^{-1} \times M^{1/2}\,L^{-3/2}$

Step 3: $[\alpha] = M^{1/2}\,L^{-1/2}\,T^{-1}$

Final: $[\beta] = M^{1/2}\,L^{-3/2}$; $[\alpha] = M^{1/2}\,L^{-1/2}\,T^{-1}$

Pressure with a log

P = A log_e(B/t − C V). Find [A B] and [B C].

Solution: Pressure with a log

Idea: log is a pure number, so [A] = [P]. Inside, each term is dimensionless.

Step 1: $[A] = [P] = M\,L^{-1}\,T^{-2}$. $B/t = 1 \Rightarrow [B] = T$

Step 2: $C\,V = 1 \Rightarrow [C] = 1/[V] = L^{-3}$

Step 3: $[A\,B] = M\,L^{-1}\,T^{-1}$; $[B\,C] = T\,L^{-3}$

Final: $[AB] = M\,L^{-1}\,T^{-1}$; $[BC] = L^{-3}\,T$

Rough work / extra work

NCERT: which displacement is wrong?

Which relation for displacement y is dimensionally NOT correct?

1. $y = a \sin(2\pi t / T)$
2. $y = a \sin(v t / \lambda)$
3. $y = \sqrt{2}\, a \sin(2\pi t / T) \cos(2\pi x / \lambda)$
4. $y = (x/t) \sin(\lambda t)$

Solution: NCERT: which displacement is wrong?

Check the angle inside sin. It must be dimensionless.

Step 1: Options 1–3: t/T, vt/λ, x/λ are all dimensionless. Fine.

Step 2: Option 4: λ t has dimensions L × T. Not an angle. Also x/t is velocity, not length.

Final answer: option 4 — $y = (x/t)\sin(\lambda t)$

New fundamental quantities

Sometimes the question says: treat force, acceleration and time as the basic set. Then rewrite length or energy using only those letters. You assume length = F^α a^β t^γ, put dimensions of both sides, and match powers of M, L, T.

A fast board shortcut: start from a formula you already trust. From $s = (1/2)\, a\, t^2$ you see length looks like $a\, t^2$. Force does not appear, so $F^0\, a^1\, t^2$.

Assume powers → equate M, L, T → solve α, β, γ

Length in terms of F, a, t

If F, a and t are fundamental, dimension of length is

1. $F^0\ a^1\ T^2$
2. $F\ a^2\ T^2$
3. $F\ a^2\ T^0$
4. $F^0\ a\ T$

Solution: Length in terms of F, a, t

Shortcut: length from kinematics looks like a t^2.

Step 1: $s \propto a\,t^2 \Rightarrow [\text{length}] = F^0\,a^1\,T^2$

Step 2: Full method: L = F^α a^β t^γ gives α = 0, β = 1, γ = 2.

Final answer: $F^0\,a^1\,T^2$

Rough work / extra work

Energy in terms of F, a, t

If force, acceleration and time are basic, dimension of energy is

1. $F^2 A^{-1} T$
2. $F A T^2$
3. $F A T^{-2}$
4. $F A^{-1} T$

Solution: Energy in terms of F, a, t

Energy = force × distance, and distance ~ a t^2.

Step 1: $E = F \times l = F \times (a\ t^2) = F\ A\ T^2$

Final answer: $F\ A\ T^2$

Surface tension from E, V, T

If energy E, velocity V and time T are fundamental, surface tension is

1. $[E\ V^{-2}\ T^{-2}]$
2. $[E^{-2}\ V\ T^{-2}]$
3. $[E^{-2}\ V^{-2}\ T^{2}]$
4. $[E\ V^{-2}\ T^{2}]$

Solution: Surface tension from E, V, T

Surface tension S = F/l. Rewrite using energy and length = V T.

Step 1: $S = F/l = (F\,l)/l^2 = E / l^2$

Step 2: $l = V\,T \Rightarrow S = E / (V\,T)^2 = E\,V^{-2}\,T^{-2}$

Final answer: $[E\,V^{-2}\,T^{-2}]$

Deducing a relation by powers

Suppose critical velocity depends on viscosity η, density ρ and radius r. Write $v_c = \eta^x \rho^y r^z$. Put dimensions of every letter. Match powers of M, L and T. Solve the three equations for x, y, z.

Board tip: often attack the mass power first. It is usually the simplest equation.

$Q = k A^a B^b C^c$ → equate [M], [L], [T]

Rough work / extra work

Critical velocity exponents

If $v_c = [\eta^x \rho^y r^z]$, find x, y, z. (AIPMT style)

1. 1, 1, 1
2. 1, −1, −1
3. −1, −1, 1
4. −1, 1, −1

Solution: Critical velocity exponents

Need [η] first. From $F = 6\pi \eta r v$, $[\eta] = M L^{-1} T^{-1}$.

Step 1: $L T^{-1} = (M L^{-1} T^{-1})^{x} (M L^{-3})^{y} (L)^{z}$

Step 2: Time: $-1 = -x \Rightarrow x = 1$. Mass: $0 = x+y \Rightarrow y = -1$.

Step 3: Length: $1 = -x - 3y + z \Rightarrow z = -1$.

Final answer: $x = 1, y = -1, z = -1$

Spring frequency powers

$f = c\, m^x K^y$. Find x and y. (c dimensionless)

1. $x = 1/2,\ y = 1/2$
2. $x = -1/2,\ y = -1/2$
3. $x = 1/2,\ y = -1/2$
4. $x = -1/2,\ y = 1/2$

Solution: Spring frequency powers

$[f] = T^{-1}$. From $F = K x$, $[K] = M T^{-2}$.

Step 1: $T^{-1} = M^{x} (M T^{-2})^{y} \Rightarrow -1 = -2y \Rightarrow y = 1/2$

Step 2: Mass: $0 = x + y \Rightarrow x = -1/2$

Final answer: $x = -1/2$, $y = 1/2$

NEET: length from h, G, c

Which combination of h, G, c has dimensions of length? [NEET]

1. $\sqrt{(h\,G) / c^{3/2}}$
2. $\sqrt{(h\,G) / c^{5/2}}$
3. $\sqrt{(h\,c / G)}$
4. $\sqrt{(G\,c) / h^{3/2}}$

Solution: NEET: length from h, G, c

Tip: first match mass. [h] = $M L^2 T^{-1}$, [G] = $M^{-1} L^3 T^{-2}$.

Step 1: $\sqrt{(h G)}$ kills mass. Remaining length and time need $c^{3/2}$.

Step 2: Check: $\sqrt{(h G)}/c^{3/2} \rightarrow L^1$. That is option 1.

Final answer: $\sqrt{(h G)} / c^{3/2}$

Rough work / extra work

Four kinds of quantities (board map)

Before significant figures, lock this language. A quantity may or may not have dimensions, and it may or may not change from problem to problem.

Dimensional variable

Has dimensions. Value can change. Example: force, velocity.

Dimensionless variable

No dimensions. Value can change. Example: strain, angle in a problem.

Dimensional constant

Has dimensions. Fixed value. Example: G, h, c.

Dimensionless constant

No dimensions. Fixed number. Example: π, e, 2 in formulas.

Which is a dimensional constant?

Which of the following is a dimensional constant?

1. Relative density
2. Gravitational constant
3. Refractive index
4. Poisson's ratio

Solution: Which is a dimensional constant?

Relative density, μ, Poisson's ratio are ratios — dimensionless.

Step 1: G has a fixed value and dimensions $M^{-1} L^{3} T^{-2}$.

Final answer: gravitational constant

Limitations of dimensional analysis

Dimensions are powerful, but they are not magic. They cannot invent a pure number like 2π. They cannot invent sin or cos. They struggle when two different quantities share the same dimensions. And they cannot build a formula that is a sum of two different kinds of terms.

1. Cannot find dimensionless constants (2π, 1/2, e).
2. Cannot derive trig / exponential pieces of a formula.
3. Fails if the quantity depends on more than three independent variables (with only M, L, T).
4. Cannot derive formulas that need + or − of unlike structures (like $s = ut + \frac{1}{2}at^2$).

Dimensionally possible does not mean physically true.

Rough work / extra work

Which equation can be derived dimensionally?

Which of the following can be derived using dimensions alone?

1. $s = v\,t - (1/2)\,a\,t^2$
2. $v^2 = u^2 - 2\,a\,s$
3. $h = \omega^2\,r^2 / (2\,g)$
4. $v = d / t$

Solution: Which equation can be derived dimensionally?

Sums with two terms, or unknown numbers like 1/2 and 2, cannot be fixed by dimensions.

Step 1: v = d/t is a single product/quotient. Dimensions can support it.

Best pick among these: v = d/t (option 4)

Significant figures — why they exist

A measurement is never infinitely precise. Significant figures tell us which digits in a reported number are trustworthy. They are the digits that carry real information from the measuring instrument.

Example idea: if a scale can only read to 0.1 cm, writing 12.345 cm pretends we know more than we do. Significant-figure rules keep our writing honest.

Significant digits = the digits that count in a measured value

Rules to count significant digits

1. Every non-zero digit is significant. (234 → 3)
2. Zeros trapped between non-zero digits are significant. (402 → 3)
3. Trailing zeros with no decimal point are not significant. (5280 → 3)
4. Trailing zeros after a decimal point are significant. (23.500 → 5)
5. Leading zeros before the first non-zero digit are not significant. (0.034 → 2)
6. Exact counted numbers have infinite significant figures. (12 eggs)
7. In $a \times 10^n$ form, only the digits in a are significant. (1.608×10^{23} → 4)

The power of ten itself is not a significant digit.

Counting table (board examples)

Value	Significant digits	Why
23.500 m	5	zeros after decimal count
402 cm	3	zero between non-zeros
5280 kg	3	no decimal — trailing zero drops
6.780	4	trailing zero after decimal
0.034 g	2	leading zeros do not count
3.070 g	4	middle + trailing after decimal
1.608×10^{23}	4	only the mantissa

Rough work / extra work

Count the significant digits

How many significant digits are in 0.0340?

1. 2
2. 3
3. 4
4. 5

Solution: Count the significant digits

Leading zeros before 3 do not count. Trailing zero after decimal does.

Step 1: Digits that count: 3, 4, 0 → three significant digits.

Final answer: 3

Scientific notation count

Number of significant digits in 2.50×10^{-4} is

1. 1
2. 2
3. 3
4. 4

Solution: Scientific notation count

Look only at 2.50. The power −4 is not counted.

Step 1: 2, 5 and trailing 0 after decimal → three significant digits.

Final answer: 3

Units 03 lock lines

Inside sin / log / exp → dimensionless. Outside factor matches left side.

New fundamentals: assume powers, equate M L T, or use a known formula shortcut.

Dimensional constant example: G. Dimensionless constant example: π.

Limit: cannot get pure numbers or invent +/− multi-term formulas.

Significant figures: count only the trustworthy digits of a measurement.

Rough work / extra work

Ready theory lines

1. Arguments of trigonometric, logarithmic and exponential functions are dimensionless.

2. Dimensional analysis can check a formula, convert units, and find powers in a product law.

3. It cannot determine dimensionless constants or formulas with + / − of unlike terms.

4. Significant figures are the meaningful digits in a measured number.

5. Leading zeros are never significant; trailing zeros after a decimal are.

For 3 marks: definition + one use + one limitation of dimensional analysis.

Practice

1. If $y = A\, e^{Bx} + C't$, find [A], [B] and [C].

2. If F, A, D are fundamental, write Young's modulus in that set.

3. Write five limitations of dimensional analysis.

4. Count significant digits: 0.002040 and 2.040×10^3.

5. Which is dimensionless: G, μ, strain, Planck's constant?

Next: more on errors and measuring instruments (later PW lectures).